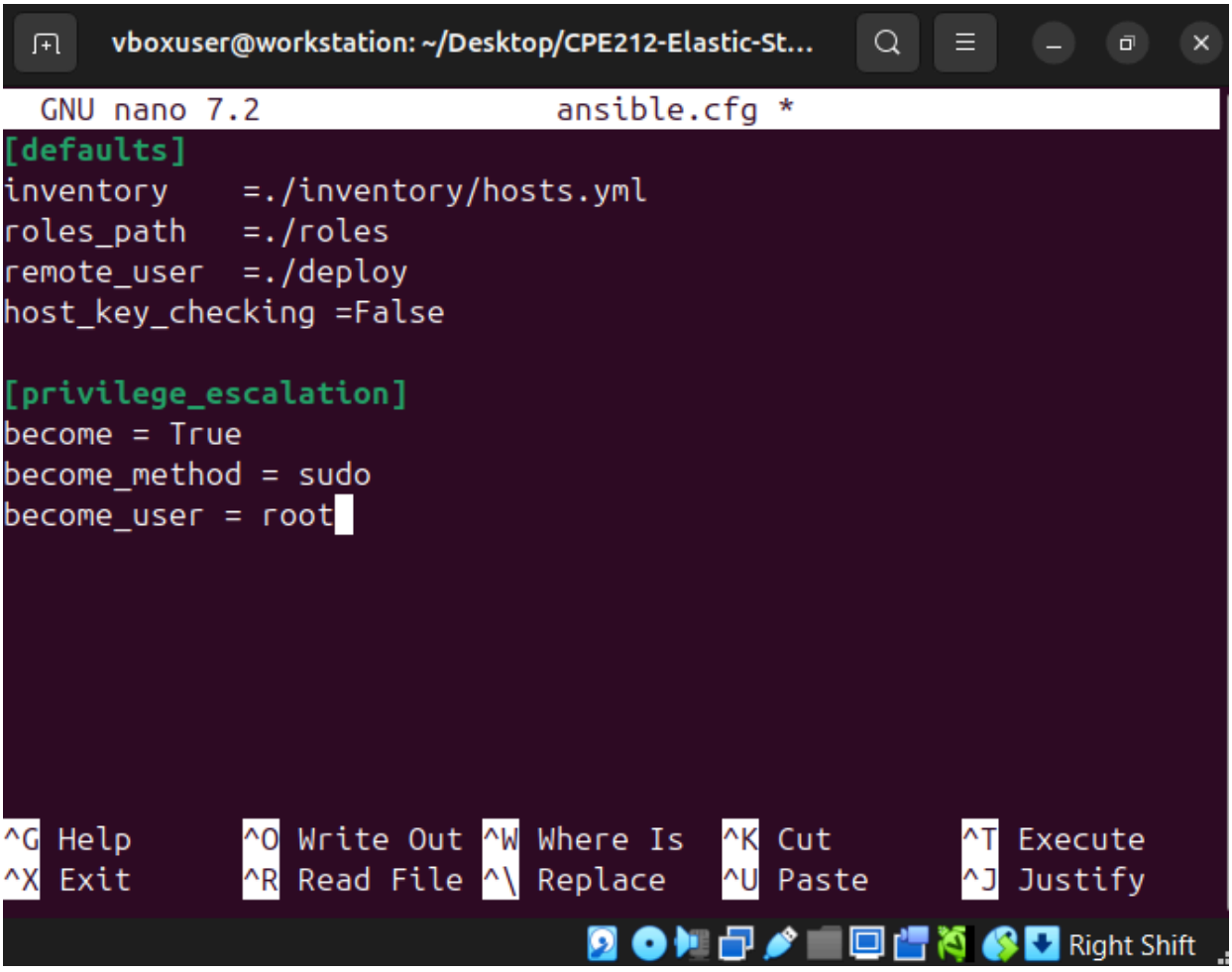
COMPUTER ENGINEERING
QUEZON CITY

COURSE CODE/TITLE:	CPE 212 - Automating Server Management	SECTION:	CPE31S3
DATE PERFORMED:	October 1, 2026	DATE SUBMITTED:	October 1, 2026
NAME:	Nadine A. Malabanan	INSTRUCTOR:	Engr. Jimlord Queado

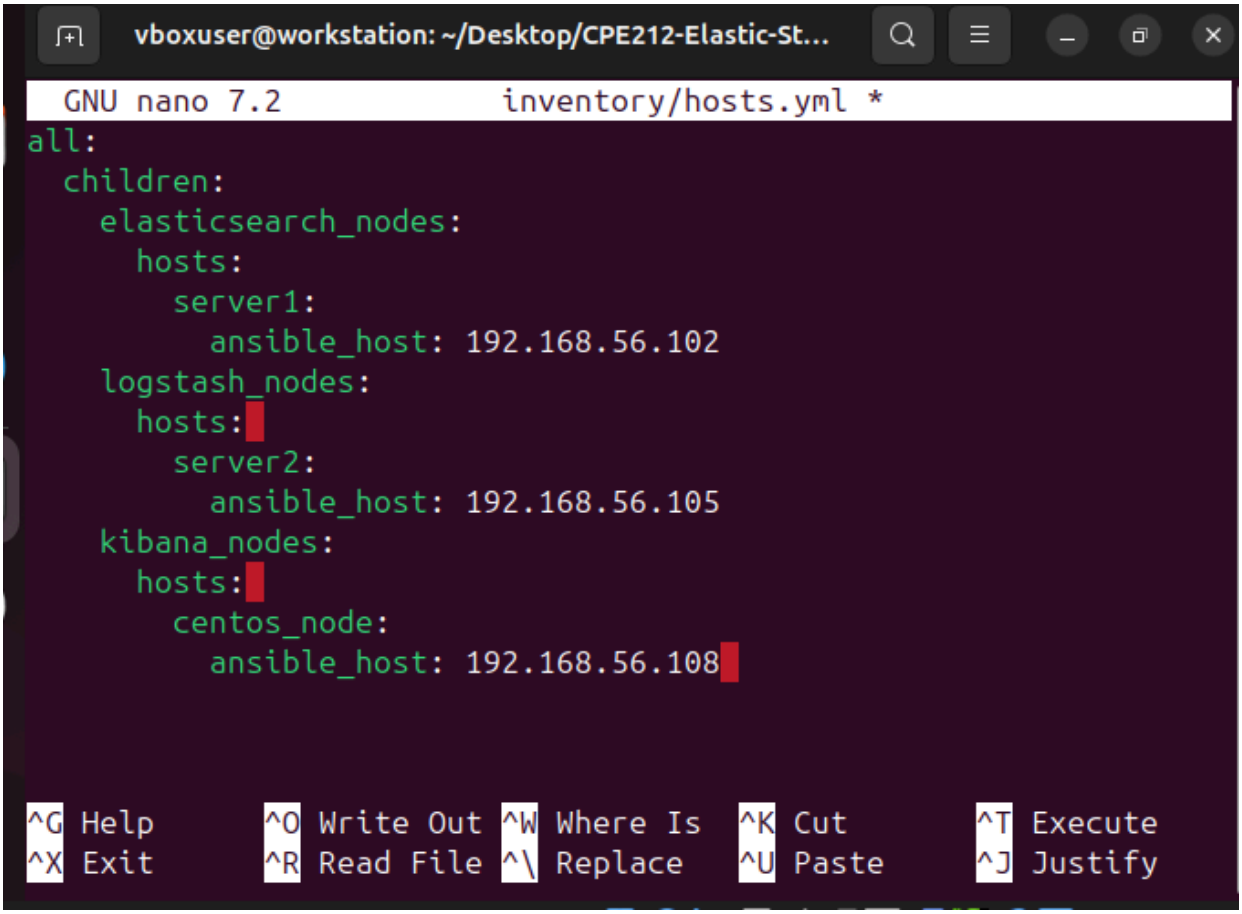
ACTIVITY NO. ___ [ACTIVITY NAME]

1. PROCEDURE OUTPUT

What procedures did you perform?



nano ansible.cfg



nano inventory/hosts.yml

```
vboxuser@workstation:~/Desktop/CPE212-Elastic-Stack-Ansible/elasti
c-stack-ansible$ mkdir -p group_vars
cat << 'EOF' > group_vars/all.yml
---
es_host: "192.168.56.102"
es_port: 9200
kb_port: 5601
EOF
```

nano inventory/group_vars/all.yml

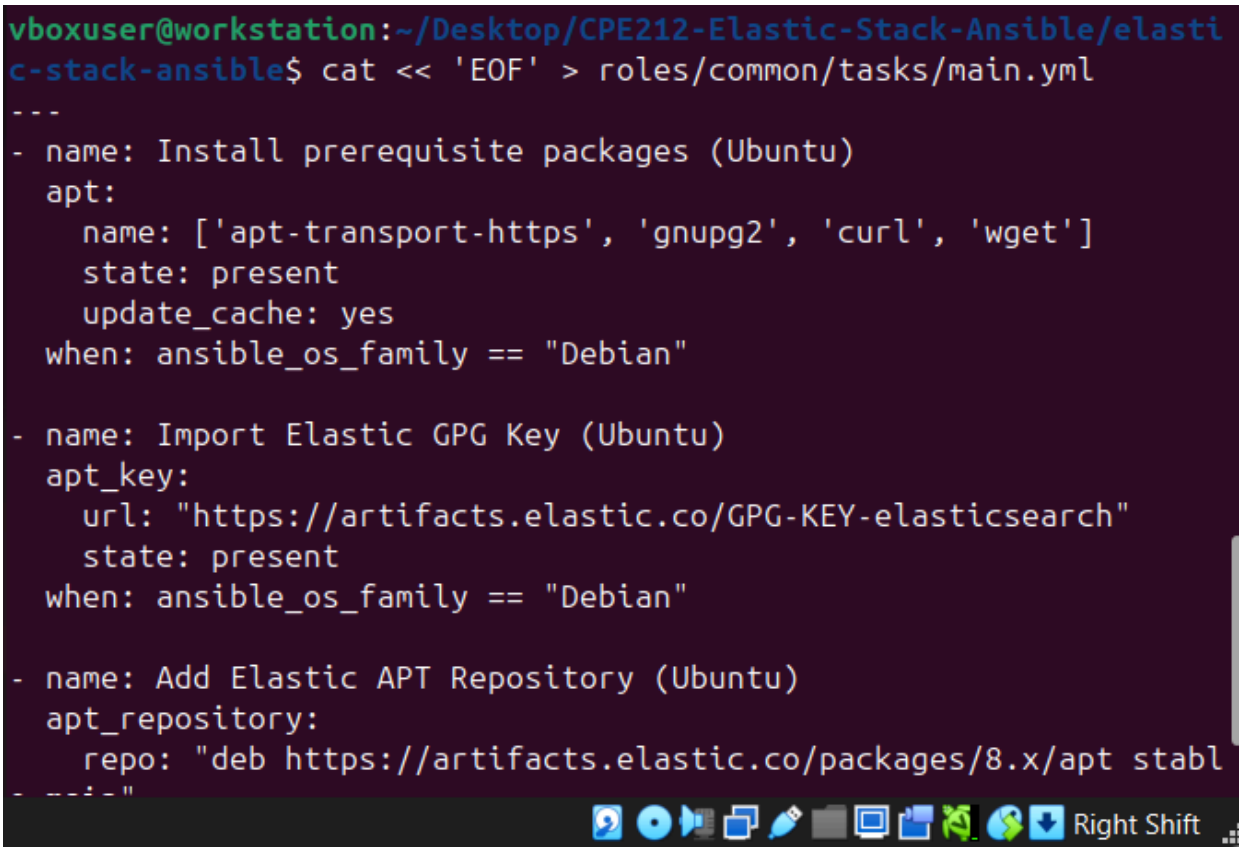
```
---
- name: Install Elasticsearch
  package:
    name: elasticsearch
    state: present

- name: Deploy elasticsearch.yml configuration
  template:
    src: elasticsearch.yml.j2
    dest: /etc/elasticsearch/elasticsearch.yml
    owner: root
    group: elasticsearch
    mode: '0660'

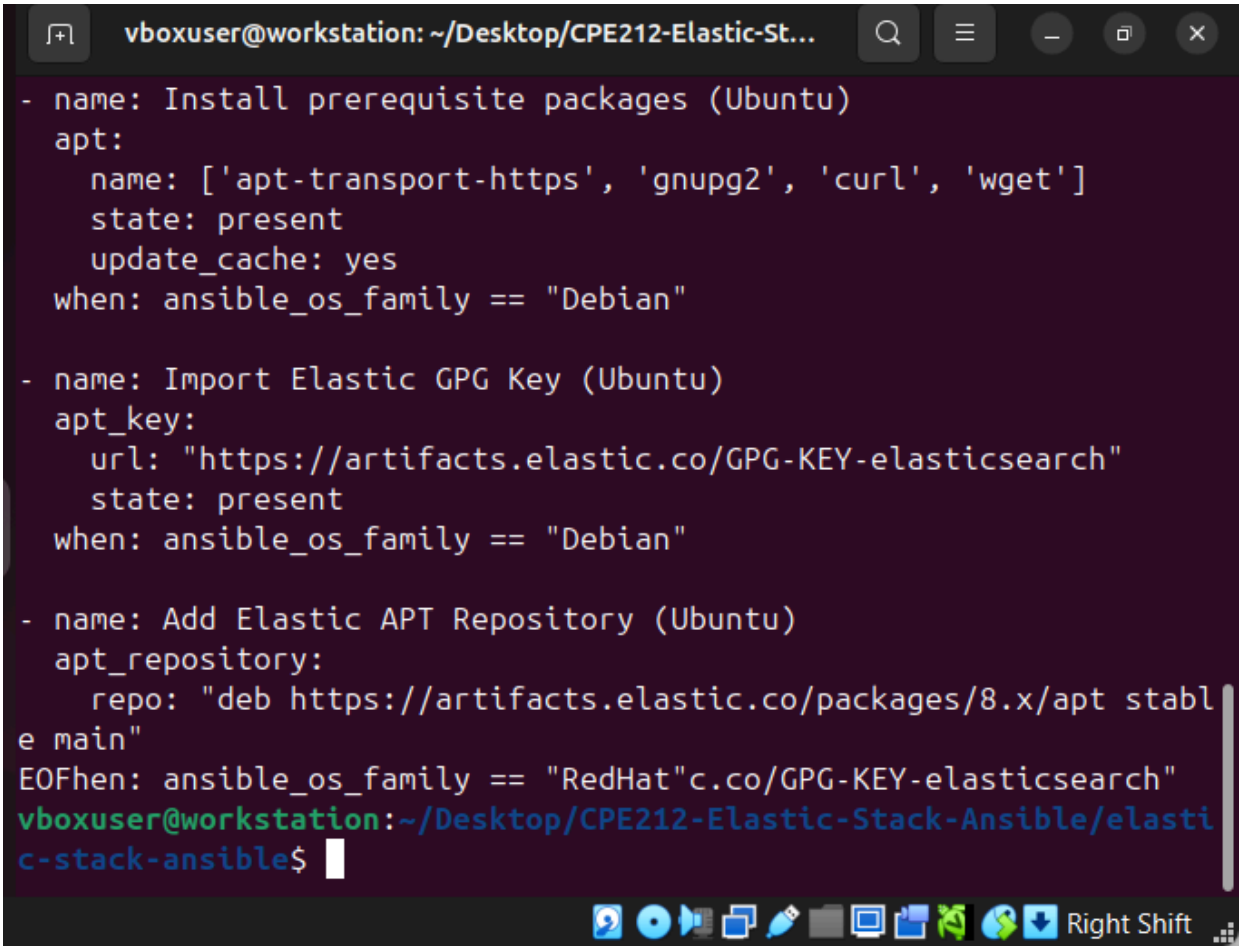
- name: Fix permissions on /var/lib/elasticsearch and /var/log/ela
sticsearch
  file:
    path: "{{ item }}"
    owner: elasticsearch
    group: elasticsearch
    state: directory
    recurse: yes
  loop:
    - /var/lib/elasticsearch
    - /var/log/elasticsearch

- name: Enable and restart Elasticsearch service
  systemd:
    name: elasticsearch
    state: restarted
    daemon_reload: yes
    enabled: yes
EOF
```

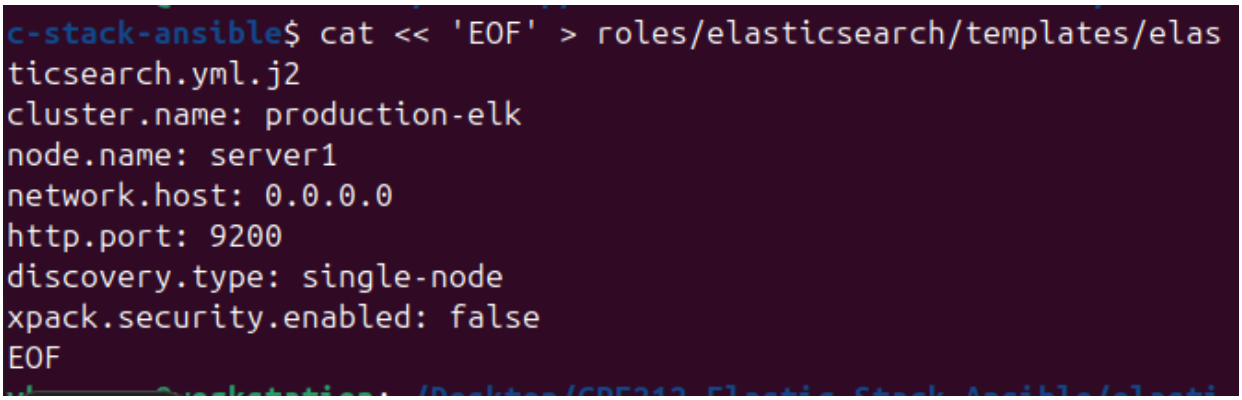
Elastic main tasks


vboxuser@workstation:~/Desktop/CPE212-Elastic-Stack-Ansible/elasti
c-stack-ansible$ cat << 'EOF' > roles/common/tasks/main.yml

- name: Install prerequisite packages (Ubuntu)
  apt:
    name: ['apt-transport-https', 'gnupg2', 'curl', 'wget']
    state: present
    update_cache: yes
  when: ansible_os_family == "Debian"
- name: Import Elastic GPG Key (Ubuntu)
  apt_key:
    url: "https://artifacts.elastic.co/GPG-KEY-elasticsearch"
    state: present
  when: ansible_os_family == "Debian"
- name: Add Elastic APT Repository (Ubuntu)
  apt_repository:
    repo: "deb https://artifacts.elastic.co/packages/8.x/apt stabl
Right Shift



Elasticsearch template



Elastic config template

```
c-stack-ansible$ cat << 'EOF' > roles/elasticsearch/tasks/main.yml

---
- name: Install Elasticsearch
  package:
    name: elasticsearch
    state: present

- name: Set Elasticsearch JVM Heap Size (512m)
  copy:
    dest: /etc/elasticsearch/jvm.options.d/heap.options
    content: |
      -Xms512m
      -Xmx512m
    owner: root
    group: elasticsearch
    mode: '0660'

- name: Deploy elasticsearch.yml configuration
EOF enabled: yesedearcht Elasticsearch service
```

Terminal

```
c-stack-ansible$ cat << 'EOF' > roles/elasticsearch/templates/elas
ticsearch.yml.j2
cluster.name: production-elk
node.name: {{ inventory_hostname }}
network.host: "0.0.0.0"
http.port: 9200
discovery.type: single-node
xpack.security.enabled: false
EOF
vboxuser@workstation:~/Desktop/CPE212-Elastic-Stack-Ansible/elasti
c-stack-ansible$
```

Right Shift

Elastic Search Tasks

```
vboxuser@workstation:~/Desktop/CPE212-Elastic-Stack-Ansible/elasti
c-stack-ansible$ cat << 'EOF' > roles/elasticsearch/tasks/main.yml

---
- name: Install Elasticsearch
  package:
    name: elasticsearch
    state: present

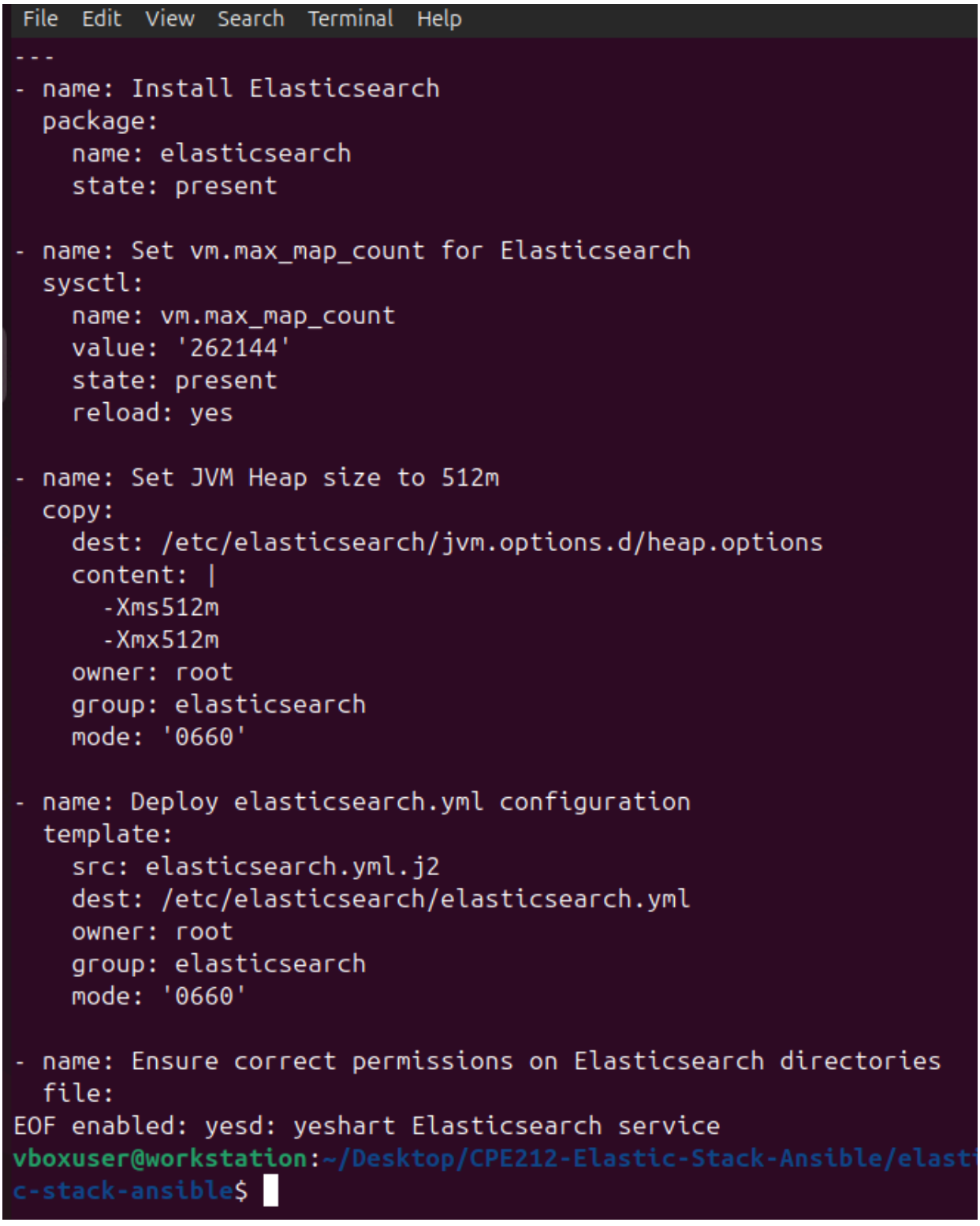
- name: Set vm.max_map_count for Elasticsearch
  sysctl:
    name: vm.max_map_count
    value: '262144'
    state: present
    load: yes

- name: Set JVM Heap size to 512m
  copy:
    dest: /etc/elasticsearch/jvm.options.d/heap.options
    content: |
      -Xms512m
      -Xmx512m
    owner: root
    group: elasticsearch
    mode: '0660'

- name: Deploy elasticsearch.yml configuration
  template:
    src: elasticsearch.yml.j2
    dest: /etc/elasticsearch/elasticsearch.yml
    owner: root
    group: elasticsearch
    mode: '0660'

- name: Ensure correct permissions on Elasticsearch directories
```

Trash



vm.max_map_count

```
c-stack-ansible$ cat << 'EOF' > roles/elasticsearch/tasks/main.yml

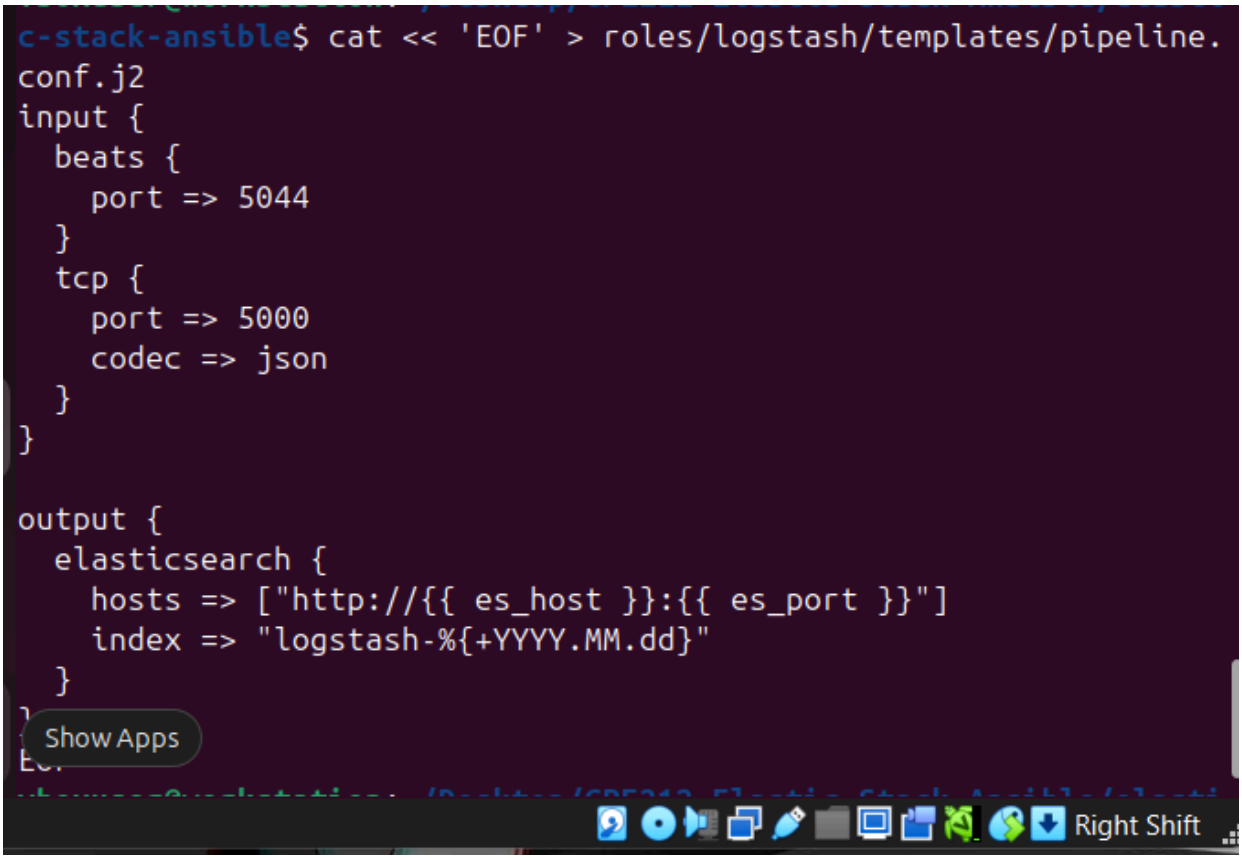
---
- name: Install Elasticsearch
  package:
    name: elasticsearch
    state: present

- name: Deploy elasticsearch.yml configuration
  template:
    src: elasticsearch.yml.j2
    dest: /etc/elasticsearch/elasticsearch.yml
    owner: root
    group: elasticsearch
    mode: '0660'

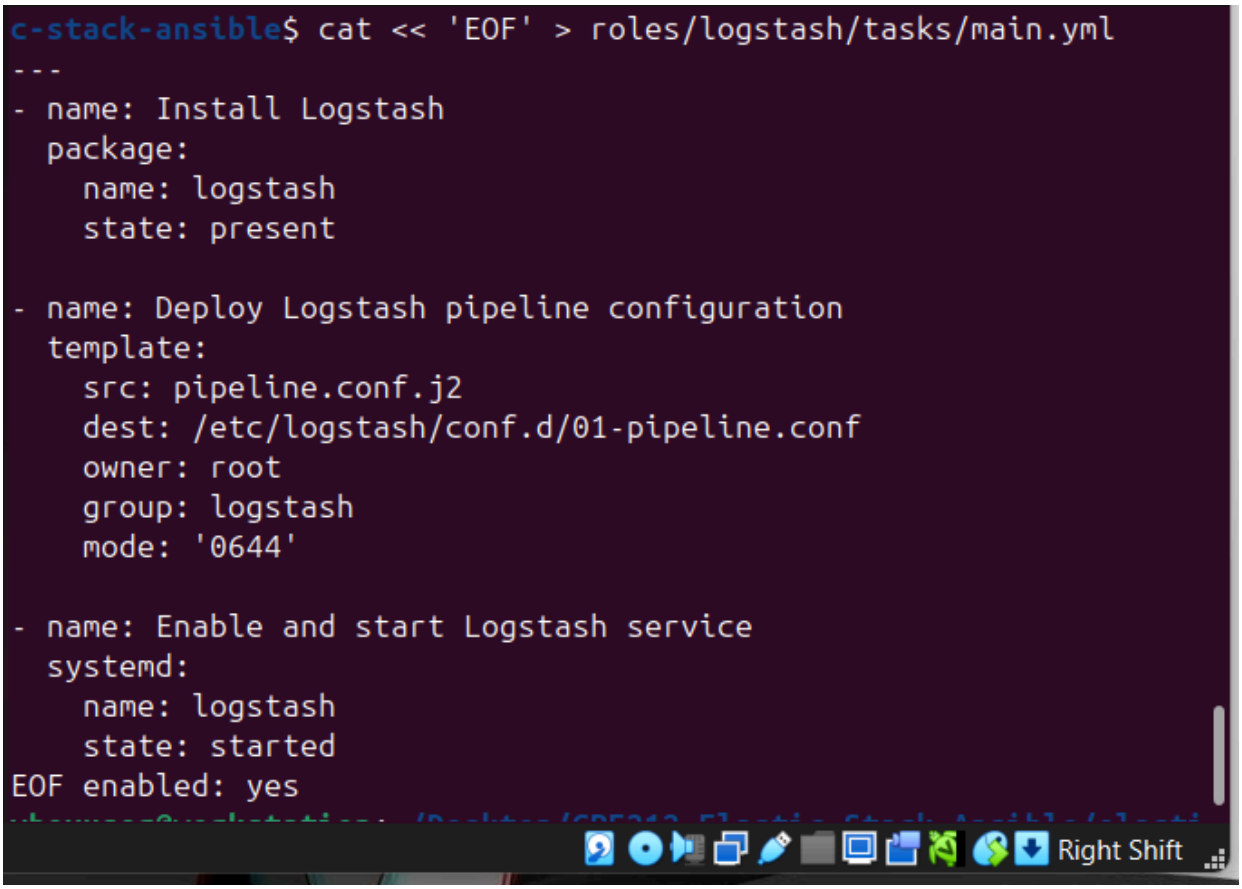
- name: Enable and start Elasticsearch service
  systemd:
    name: elasticsearch
EOF enabled: yesed
```



Logstash Role Setup



Logstash Tasks


c-stack-ansible$ cat << 'EOF' > roles/logstash/tasks/main.yml
---
- name: Install Logstash
  package:
    name: logstash
    state: present

- name: Deploy Logstash pipeline configuration
  template:
    src: pipeline.conf.j2
    dest: /etc/logstash/conf.d/01-pipeline.conf
    owner: root
    group: logstash
    mode: '0644'

- name: Enable and start Logstash service
  systemd:
    name: logstash
    state: started
EOF enabled: yes
Right Shift

```
vboxuser@workstation:~/Desktop/CPE212-Elastic-Stack-Ansible/elasti
c-stack-ansible$ cat << 'EOF' > roles/logstash/templates/pipeline.
conf.j2
input {
  beats {
    port => 5044
  }
  tcp {
    port => 5000
    codec => json
  }
}

output {
  elasticsearch {
    hosts => ["http://192.168.56.102:9200"]
    index => "logstash-%{+YYYY.MM.dd}"
  }
}
EOF
```

Logstash template

Kibana Role Setup

```
vboxuser@workstation:~/Desktop/CPE212-Elastic-Stack-Ansible/elasti
c-stack-ansible$ cat << 'EOF' > roles/kibana/templates/kibana.yml.
j2
server.port: {{ kb_port }}
server.host: "0.0.0.0"
elasticsearch.hosts: ["http://{{ es_host }}:{{ es_port }}"]
EOF
```

Kibana Tasks

```
c-stack-ansible$ cat << 'EOF' > roles/kibana/tasks/main.yml
---
- name: Install Kibana
  package:
    name: kibana
    state: present

- name: Deploy kibana.yml configuration
  template:
    src: kibana.yml.j2
    dest: /etc/kibana/kibana.yml
    owner: root
    group: kibana
    mode: '0660'

- name: Enable and start Kibana service
  systemd:
    name: kibana
    state: started
EOF enabled: yes
```

Right Shift

Kibana template

```
vboxuser@workstation:~/Desktop/CPE212-Elastic-Stack-Ansible/elasti
c-stack-ansible$ cat << 'EOF' > roles/kibana/templates/kibana.yml.
j2
server.port: 5601
server.host: "0.0.0.0"
elasticsearch.hosts: ["http://192.168.56.102:9200"]
EOF
vboxuser@workstation:~/Desktop/CPE212-Elastic-Stack-Ansible/elasti
```

playbook.yml

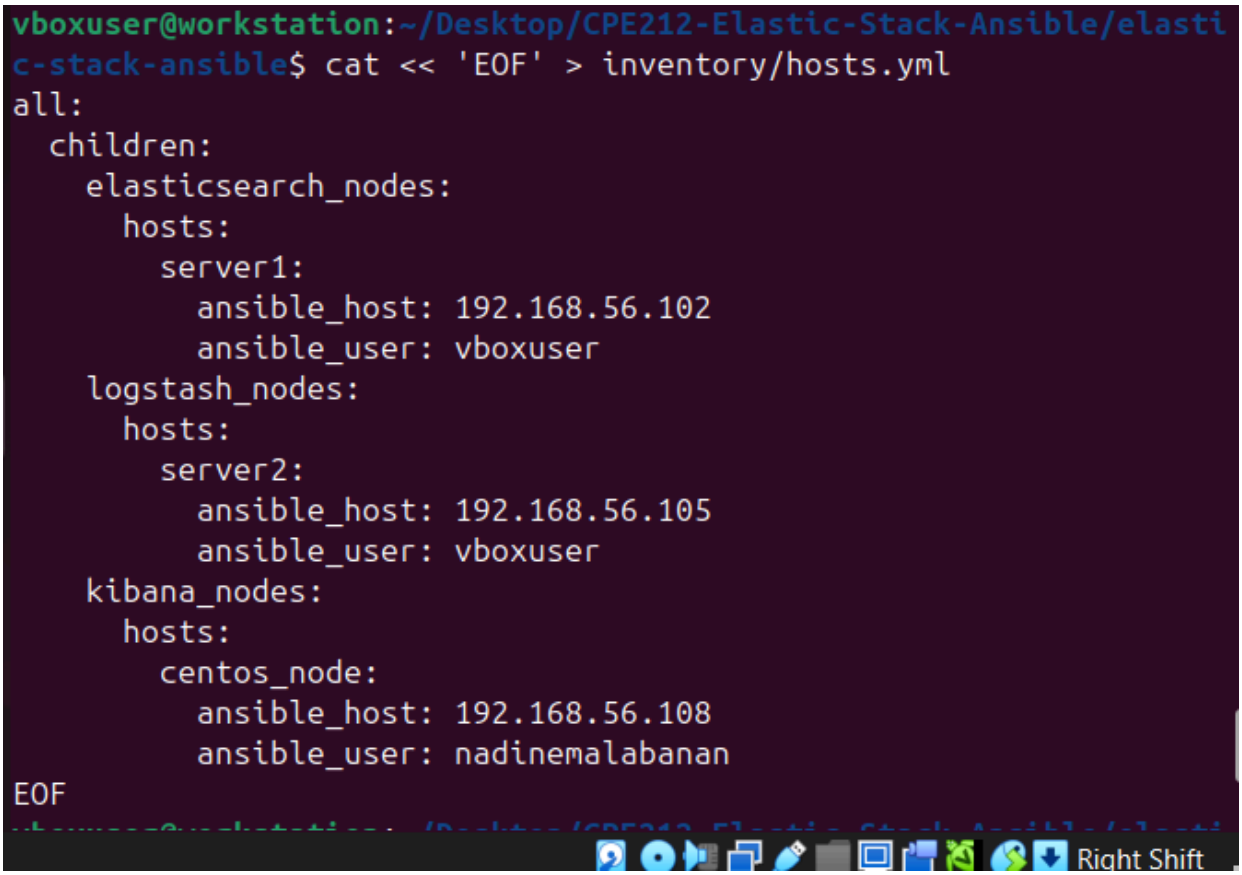
```
vboxuser@workstation:~/Desktop/CPE212-Elastic-Stack-Ansible/elasti
c-stack-ansible$ cat << 'EOF' > playbook.yml
---
- name: Apply Common Base Configuration
  hosts: all
  become: yes
  roles:
    - common

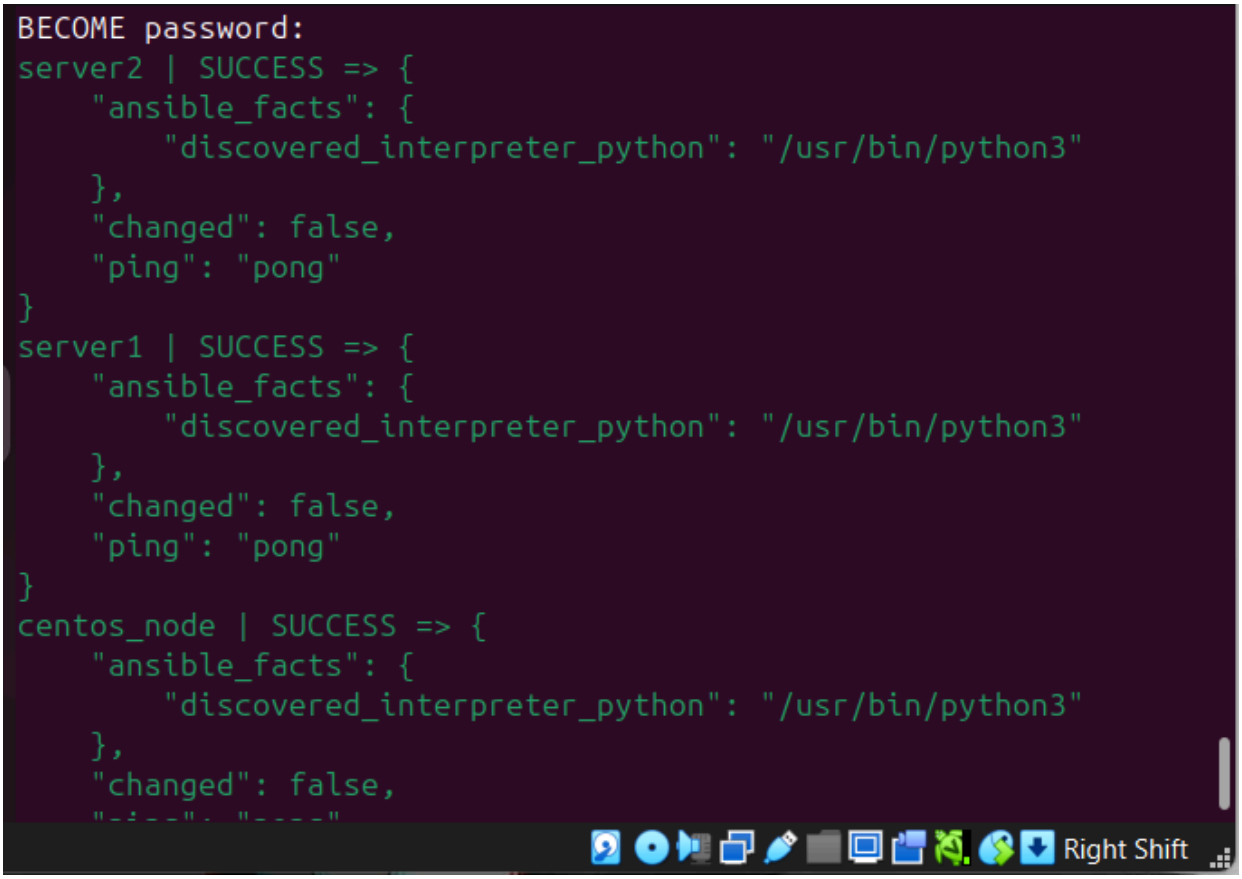
- name: Deploy Elasticsearch Node
  hosts: elasticsearch_nodes
  become: yes
  roles:
    - elasticsearch

- name: Deploy Logstash Node
  hosts: logstash_nodes
  become: yes
```

Show Apps

inventory/hosts.yml


vboxuser@workstation:~/Desktop/CPE212-Elastic-Stack-Ansible/elasti
c-stack-ansible$ cat << 'EOF' > inventory/hosts.yml
all:
children:
elasticsearch_nodes:
hosts:
server1:
ansible_host: 192.168.56.102
ansible_user: vboxuser
logstash_nodes:
hosts:
server2:
ansible_host: 192.168.56.105
ansible_user: vboxuser
kibana_nodes:
hosts:
centos_node:
ansible_host: 192.168.56.108
ansible_user: nadinemalabanan
EOF
Right Shift



permissions fix at systemd reload sa Tasks

Output:

```
vboxuser@workstation:~/Desktop/CPE212-Elastic-Stack-Ansible/elastic-stack-ansible$ ansible-playbook -i inventory/hosts.yml playbook.yml -K
BECOME password:

PLAY [Apply Common Base Configuration] ******************************************

TASK [Gathering Facts] **********************************************************
ok: [server1]
ok: [server2]
ok: [centos_node]

TASK [common : Install prerequisite packages (Ubuntu)] **************************
skipping: [centos_node]
ok: [server1]
ok: [server2]

TASK [common : Import Elastic GPG Key (Ubuntu)] *********************************
skipping: [centos_node]
ok: [server1]
ok: [server2]

TASK [common : Add Elastic APT Repository (Ubuntu)] *****************************
skipping: [centos_node]
ok: [server1]
ok: [server2]

TASK [common : Import Elastic GPG Key (CentOS)] *********************************
skipping: [server1]
```

```
TASK [common : Import Elastic GPG Key (CentOS)] ******************
**************
skipping: [server1]
skipping: [server2]
ok: [centos_node]

TASK [common : Add Elastic YUM Repository (CentOS)] **************
**************
skipping: [server1]
skipping: [server2]
ok: [centos_node]

PLAY [Deploy Elasticsearch Node] *********************************
**************

TASK [Gathering Facts] *******************************************
**************
ok: [server1]

TASK [elasticsearch : Install Elasticsearch] *********************
**************
ok: [server1]

TASK [elasticsearch : Set vm.max_map_count for Elasticsearch] ****
**************
ok: [server1]

TASK [elasticsearch : Set JVM Heap size to 512m] *****************
**************
changed: [server1]

TASK [elasticsearch : Deploy elasticsearch.yml configuration] ****
**************
changed: [server1]
```

```
TASK [elasticsearch : Clear old node states and temp locks] ******
**************
changed: [server1] => (item=/var/lib/elasticsearch/nodes)
ok: [server1] => (item=/tmp/elasticsearch*)

TASK [elasticsearch : Set systemd package ownership on binaries (/
usr/share/elasticsearch)] ***
ok: [server1]

TASK [elasticsearch : Set elasticsearch ownership on data, log, an
d config paths] ***
ok: [server1] => (item=/var/lib/elasticsearch)
changed: [server1] => (item=/var/log/elasticsearch)
changed: [server1] => (item=/etc/elasticsearch)

TASK [elasticsearch : Enable and restart Elasticsearch service] **
**************
changed: [server1]

PLAY [Deploy Logstash Node] ***************************************
**************

TASK [Gathering Facts] ********************************************
**************
ok: [server2]

TASK [logstash : Install Logstash] ********************************
**************
ok: [server2]

TASK [logstash : Deploy Logstash pipeline configuration] *********
**************
ok: [server2]

TASK [logstash : Enable and start Logstash service] **************
**************
```

```
**************
changed: [server1]

PLAY [Deploy Logstash Node] ************************************
**************

TASK [Gathering Facts] *****************************************
**************
ok: [server2]

TASK [logstash : Install Logstash] *****************************
**************
ok: [server2]

TASK [logstash : Deploy Logstash pipeline configuration] *********
**************
ok: [server2]

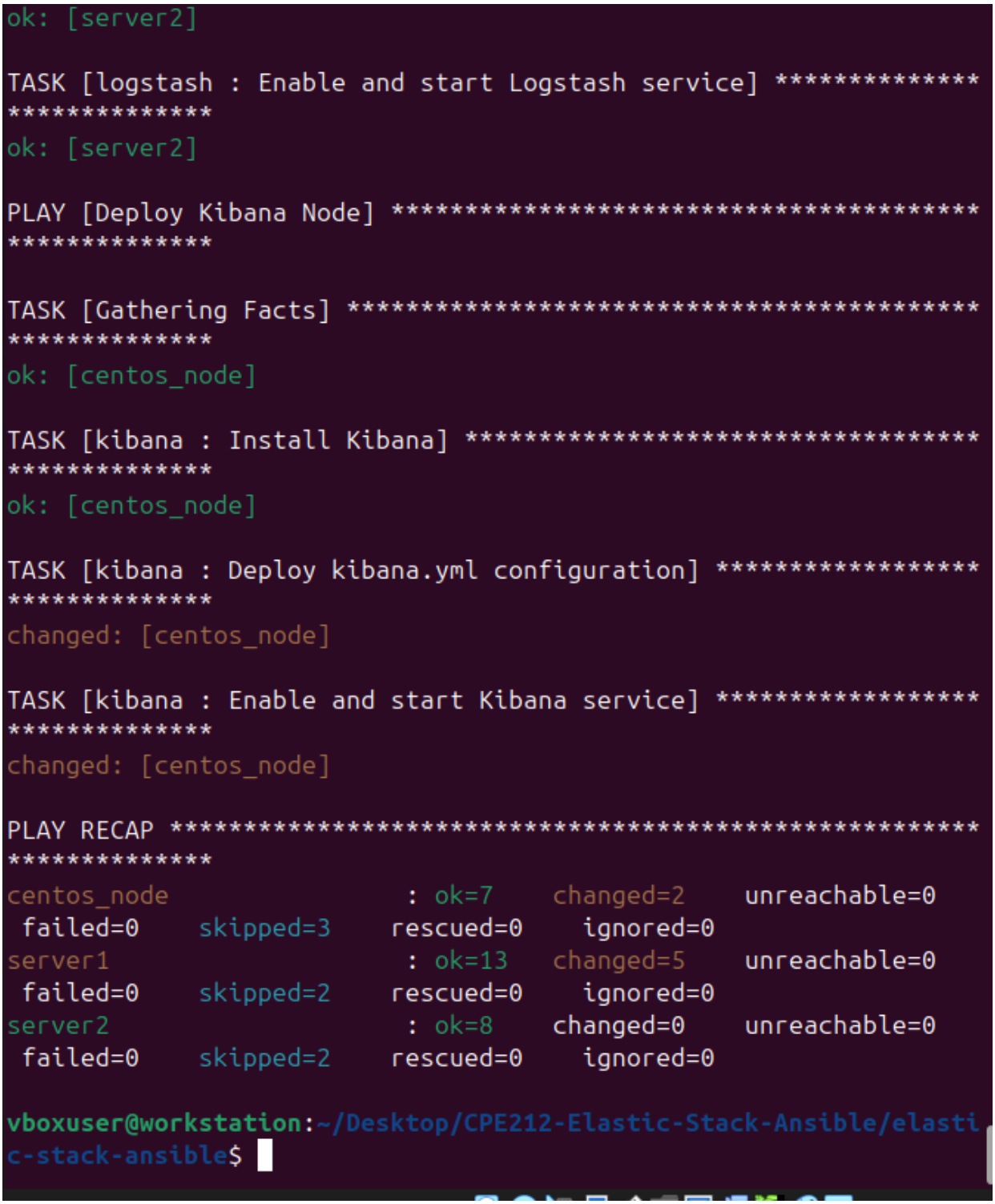
TASK [logstash : Enable and start Logstash service] **************
**************
ok: [server2]

PLAY [Deploy Kibana Node] **************************************
**************

TASK [Gathering Facts] *****************************************
**************
ok: [centos_node]

TASK [kibana : Install Kibana] *********************************
**************
ok: [centos_node]

TASK [kibana : Deploy kibana.yml configuration] ******************
**************
changed: [centos_node]
```


ok: [server2]
TASK [logstash : Enable and start Logstash service] **********
****
ok: [server2]
PLAY [Deploy Kibana Node] **
****
TASK [Gathering Facts] ***
****
ok: [centos_node]
TASK [kibana : Install Kibana] ***********************************
****
ok: [centos_node]
TASK [kibana : Deploy kibana.yml configuration] ******************
****
changed: [centos_node]
TASK [kibana : Enable and start Kibana service] ******************
****
changed: [centos_node]
PLAY RECAP ***
****
centos_node : ok=7 changed=2 unreachable=0
failed=0 skipped=3 rescued=0 ignored=0
server1 : ok=13 changed=5 unreachable=0
failed=0 skipped=2 rescued=0 ignored=0
server2 : ok=8 changed=0 unreachable=0
failed=0 skipped=2 rescued=0 ignored=0
vboxuser@workstation:~/Desktop/CPE212-Elastic-Stack-Ansible/elasti
c-stack-ansible$

Node Hostname	IP Address	OS Platform	Role/Service	Description
Server 1	192.168.56.106	Ubuntu 22.04 LTS	Elasticsearch	Data Indexing & Search Engine Backend
Server 2	192.168.56.105	Ubuntu 22.04 LTS	Logstash	Data Collection & Pipeline Processing
centos_node	192.168.56.108	CentOS Stream	Kibana	Visualization Dashboard & Management UI
workstation	192.168.56.104	Ubuntu 22.04 LTS	Control Node	Ansible Playbook Orchestrator

2. Directory Structure and Role-Based Architecture

For this activity, I used Ansible Roles to organize the deployment of the Elastic Stack. I separated Elasticsearch, Kibana, and Logstash into different roles because each service has a different purpose and will be installed on a different server.

This activity contains three main roles: Elasticsearch, Kibana, and Logstash. Each role has its own tasks and configuration files. I also created an inventory file where I placed the IP addresses of the servers.

The Ansible control node is my workstation with IP address 192.168.56.104. Server 1 with IP address 192.168.56.102 is used for Elasticsearch, Server 2 with IP address 192.168.56.105 is used for Kibana, and the CentOS machine with IP address 192.168.56.108 is used for Logstash.

Project Structure

The project is organized into the following folders:

- inventory - contains the list of servers that Ansible will manage.
- roles/elasticsearch – contains the tasks and configuration for Elasticsearch.
- roles/kibana - contains the tasks and configuration for Kibana.
- roles/logstash - contains the tasks for Logstash.
- playbook.yml - the main playbook that runs all three roles.
- ansible.cfg - contains the basic Ansible configuration.

This structure makes the project easier to manage because each service has its own set of tasks.

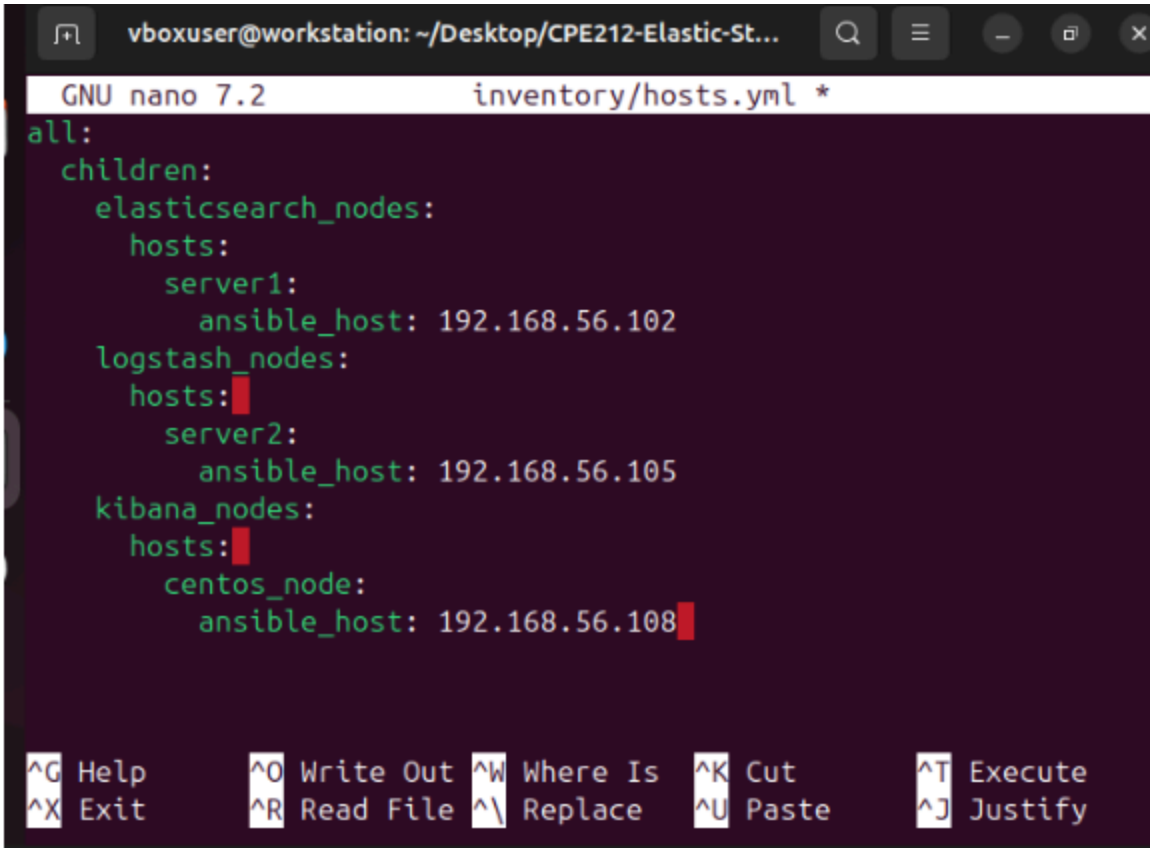
---

3. Deployment Code and Configuration

3.1 Inventory Configuration

I first created the inventory file to tell Ansible which computers it needs to connect to. I grouped the servers according to the Elastic Stack service that will be installed on each one.

The server assignments are:The workstation is only used as the Ansible control node, while the other three machines are the managed nodes.



Explanation:
The inventory allows Ansible to identify the correct server for each Elastic Stack component. This is important because Elasticsearch, Kibana, and Logstash are required to be installed on separate hosts for this activity.

3.2 Main Playbook

After creating the inventory, I created the main Ansible playbook. The purpose of the playbook is to connect the correct role to the correct server.

The Elasticsearch role is assigned to Server 1, the Kibana role is assigned to Server 2, and the Logstash role is assigned to the CentOS server.

I used become in the playbook because installing packages and modifying system configuration files requires administrator privileges.

```
vboxuser@workstation:~/Desktop/CPE212-Elastic-Stack-Ansible/elasti
c-stack-ansible$ cat << 'EOF' > playbook.yml
---
- name: Apply Common Base Configuration
  hosts: all
  become: yes
  roles:
    - common

- name: Deploy Elasticsearch Node
  hosts: elasticsearch_nodes
  become: yes
  roles:
    - elasticsearch

- name: Deploy Logstash Node
  hosts: logstash_nodes
  become: yes
```

Show Apps

Explanation:
Instead of putting all the installation tasks into one large playbook, I used roles to separate the tasks. This makes the playbook easier to understand and modify.

3.3 Elasticsearch Installation

For Elasticsearch, I used the Elasticsearch role. This role is responsible for installing Elasticsearch and configuring the service on Server 1.
The first step was installing the Elasticsearch package. After that, I configured the required system settings and adjusted the Elasticsearch memory allocation because the servers are running inside VirtualBox.
I set the Elasticsearch JVM heap to 512 MB for both the minimum and maximum heap size. This helps reduce the amount of memory used by Elasticsearch and prevents the virtual machine from becoming too slow.
I also configured the vm.max_map_count value to 262144, which is needed for Elasticsearch.
The Elasticsearch configuration was also changed so that the service could accept connections from other machines. The HTTP service uses port 9200.
Explanation:
Server 1 is responsible for running Elasticsearch. After the installation, I enabled and started the Elasticsearch service using Ansible.

3.4 Logstash Installation

For Logstash, I used the Logstash role and installed it on the CentOS machine with IP address 192.168.56.108.
The role installs the Logstash package and then enables and starts the Logstash service.
I used the systemd service manager to make sure that Logstash starts automatically and continues running after the installation.

Explanation:
This part also demonstrates that Ansible can manage different operating systems. The other Elastic Stack components are running on Ubuntu, while Logstash is running on CentOS.

3.5 Kibana Installation
For Kibana, I created a separate Kibana role and assigned it to Server 2 with IP address 192.168.56.105.
The role installs the Kibana package and places the Kibana configuration file on the server.
I configured Kibana to use port 5601. I also configured the Elasticsearch server address so that Kibana can communicate with Elasticsearch.
Explanation:
Kibana is used as the web interface of the Elastic Stack. It connects to the Elasticsearch server on Server 1.

---

4. Implementation and Troubleshooting
4.1 Memory Configuration
One of the things I considered during the installation was the memory requirement of Elasticsearch. Since I am using virtual machines, giving too much memory to Elasticsearch can affect the performance of the other machines.
To avoid this problem, I limited the Elasticsearch JVM heap to 512 MB.
I also configured the required virtual memory setting using vm.max_map_count.
After making these changes, I restarted Elasticsearch and checked its status to make sure that the service was running properly.

4.2 Permission Issues
I also checked the permissions of the Elasticsearch directories because incorrect ownership can prevent the service from starting.
The important directories include the Elasticsearch configuration, data, and log directories.
When troubleshooting, I checked the service status and reviewed the Elasticsearch logs whenever the service failed to start.
This helped me identify whether the problem was caused by permissions, configuration, or the service itself.

4.3 Elasticsearch Configuration Issue
During the configuration, I also encountered an issue related to the Elasticsearch security and SSL settings.
For this laboratory setup, I configured the security and SSL settings consistently so that Elasticsearch could start properly and communicate with Kibana.
After modifying the configuration, I restarted Elasticsearch and checked the service status again.

---

5. Deployment Verification
After creating the roles and configurations, I ran the main Ansible playbook from my workstation.
The first thing I checked was whether Ansible could connect to all three managed servers.
I used the Ansible ping module to test the connection. The expected result was a SUCCESS response from Server 1, Server 2, and the CentOS server.
After confirming the connection, I ran the main playbook to install the Elastic Stack.
The playbook executed the Elasticsearch role on Server 1, the Kibana role on Server 2, and the Logstash role on the CentOS server.
At the end of the execution, Ansible displayed a PLAY RECAP. I checked that all servers had zero unreachable and failed tasks.

5.1 Elasticsearch Verification
After the playbook finished, I checked Elasticsearch on Server 1 (192.168.56.102).

I checked the Elasticsearch service to make sure that its status was active and running.
I also tested the Elasticsearch REST API using the server's IP address and port 9200.
If the installation was successful, Elasticsearch returned information about the node, cluster name, and installed version.
This confirmed that Elasticsearch was successfully installed and running.

5.2 Kibana Verification

Next, I checked Kibana on Server 2 (192.168.56.105).
I checked the Kibana service and verified that it was running. I also checked that port 5601 was open and listening.
After that, I opened the Kibana interface using the server's IP address in a web browser.
The Kibana web interface loaded successfully, which confirmed that the Kibana service was working.

5.3 Logstash Verification

Lastly, I checked Logstash on the CentOS server (192.168.56.108).
I checked the Logstash service and confirmed that its status was active and running.
I also checked the Logstash logs to make sure there were no major errors after the installation.

---

6. Final Elastic Stack Setup

After completing the deployment, my Elastic Stack setup consisted of three separate servers.
Server 1 (192.168.56.102) runs Elasticsearch.
Server 2 (192.168.56.105) runs Kibana.
CentOS (192.168.56.108) runs Logstash.
My workstation (192.168.56.104) acts as the Ansible control node and is used to manage the three servers.
This setup allowed me to demonstrate how Ansible can be used to install and configure different services on multiple machines. It also demonstrated the use of Ansible Roles to organize the deployment.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.